



Paper ID : 260809

Subject Code: KMHC205

Roll No:

MBA

(SEM. II) THEORY EXAMINATION 2025-26

OPERATION RESEARCH FOR HOSPITAL MANAGEMENT

TIME: 3 HRS

M.MARKS: 100

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x10 = 20

(a)	Define <i>Operations Research</i> . State its two key characteristics.	CO1	K2
(b)	List any four limitations of Operations Research.	CO1	K2
(c)	What is a <i>basic feasible solution</i> in Linear Programming?	CO1	K2
(d)	State the difference between <i>North-West Corner method</i> and <i>Least Cost method</i> for transportation problems.	CO1	K2
(e)	Define <i>assignment problem</i> . What is the objective of the Hungarian method?	CO1	K2
(f)	State the concepts of queuing theory.	CO1	K2
(g)	Define <i>sequencing problem</i> . List its two assumptions.	CO1	K2
(h)	Differentiate between <i>individual replacement</i> and <i>group replacement</i> of items that fail completely.	CO1	K2
(i)	Define <i>critical path</i> in network analysis.	CO1	K2
(j)	Differentiate between <i>PERT</i> and <i>CPM</i> on the basis of time estimation.	CO1	K2

SECTION B

2. Attempt any three of the following:

10 x 3 = 30

(a)	Explain the scientific methodology of Operations Research with a step-by-step approach. Discuss the scope and importance of OR in managerial decision-making with two real-life examples.	CO1	K2
(b)	A company produces two products A and B. Each unit of A requires 2 hours of machining and 1 hour of assembly. Each unit of B requires 1 hour of machining and 2 hours of assembly. Daily available hours: machining = 100 hours, assembly = 80 hours. Profit contribution: A = ₹40/unit, B = ₹30/unit. Formulate the Linear Programming problem (do not solve graphically). Also, explain the graphical solution method for any LP problem with two variables, describe the steps.	CO1	K2
(c)	Explain the MODI method (u-v method) for finding optimal solution in a transportation problem. What is degeneracy in transportation? How is it resolved? A company has 3 factories and 4 warehouses. Initial basic feasible solution has been found. Describe the step-by-step procedure to check optimality using MODI method.	CO1	K2
(d)	Explain the Hungarian algorithm for solving an assignment problem step by step with a logical example of assigning 4 workers to 4 machines (without numerical matrix). Also, state the basic structure of a queuing system with its six components.	CO1	K2
(e)	Explain the difference between PERT and CPM on the basis of: (i) Time estimation (ii) Nature of activities (iii) Probability of completion (iv) Focus area Describe the procedure for time analysis in CPM – earliest start time (EST), earliest finish time (EFT), latest start time (LST), latest finish time (LFT), and float (slack).	CO1	K2

SECTION C

3. Attempt any one part of the following:

10 x 1 = 10

(a)	Discuss the role of Operations Research models in decision making. Classify OR models into any four types with examples for each. Why is OR considered a scientific approach to problem solving?	CO1	K2
(b)	What is the scope of Operations Research in different functional areas of management (marketing, finance, production, HR, supply chain)? Explain each with one application.	CO1	K2

4. Attempt any **one** part of the following:

10 x 1 = 10

(a)	Find the Maximum value of $Z=5X_1 + 7X_2$ Subject to the Constraints, $X_1 + X_2 \leq 4$ $3X_1 + 8X_2 \leq 24$ $10X_1 + 7X_2 \leq 35$ $X_1, X_2 \geq 0$	CO1	K2
(b)	A transportation problem has 3 sources and 4 destinations . Explain the following methods for finding initial basic feasible solution : (i) North-West Corner (NWC) method (ii) Least Cost method (iii) Vogel's Approximation Method (VAM) State which method gives the closest to optimal solution and why	CO1	K2

5. Attempt any **one** part of the following:

10 x 1 = 10

(a)	A Company has 5 jobs to be done on five machines. Any job can be done on any machine. The cost of doing the jobs on different machine are given below. Assign the jobs for different machines so as to minimize the total cost. <table><tr><th colspan="6">Machine</th></tr><tr><th>Jobs/ Machine</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr><tr><td>1</td><td>13</td><td>8</td><td>16</td><td>18</td><td>19</td></tr><tr><td>2</td><td>9</td><td>15</td><td>24</td><td>9</td><td>12</td></tr><tr><td>3</td><td>12</td><td>9</td><td>4</td><td>4</td><td>4</td></tr><tr><td>4</td><td>6</td><td>12</td><td>10</td><td>8</td><td>13</td></tr><tr><td>5</td><td>15</td><td>17</td><td>18</td><td>12</td><td>20</td></tr></table>	Machine						Jobs/ Machine	A	B	C	D	E	1	13	8	16	18	19	2	9	15	24	9	12	3	12	9	4	4	4	4	6	12	10	8	13	5	15	17	18	12	20	CO1	K2
Machine																																													
Jobs/ Machine	A	B	C	D	E																																								
1	13	8	16	18	19																																								
2	9	15	24	9	12																																								
3	12	9	4	4	4																																								
4	6	12	10	8	13																																								
5	15	17	18	12	20																																								
(b)	Explain the basic structure of a queuing system. Define the following terms with examples from a bank teller counter: <table><tr><td>(i)</td><td>Arrival rate (λ)</td><td>(ii)</td><td>Service rate (μ)</td></tr><tr><td>(iii)</td><td>Traffic intensity (ρ)</td><td>(iv)</td><td>Queue discipline</td></tr><tr><td>(v)</td><td colspan="3">Balking, Reneging, Jockeying</td></tr></table> State why Poisson distribution is used for arrival patterns and Exponential distribution for service times	(i)	Arrival rate (λ)	(ii)	Service rate (μ)	(iii)	Traffic intensity (ρ)	(iv)	Queue discipline	(v)	Balking, Reneging, Jockeying			CO1	K2																														
(i)	Arrival rate (λ)	(ii)	Service rate (μ)																																										
(iii)	Traffic intensity (ρ)	(iv)	Queue discipline																																										
(v)	Balking, Reneging, Jockeying																																												

6. Attempt any **one** part of the following:

10 x 1 = 10

(a)	Explain Johnson's algorithm for sequencing jobs through 2 machines. A factory has 5 jobs to be processed on Machine A then Machine B. Describe the step-by-step procedure to find the optimal sequence (without numerical). Also explain the assumptions of sequencing problems.	CO1	K2
(b)	A machine costs ₹1,00,000. Its annual maintenance costs increase with age as follows: Year 1: ₹5,000, Year 2: ₹8,000, Year 3: ₹12,000, Year 4: ₹18,000, Year 5: ₹25,000 Resale value declines: Year 1: ₹70,000, Year 2: ₹50,000, Year 3: ₹35,000, Year 4: ₹20,000, Year 5: ₹10,000 Determine at what time it should be replaced? Differentiate between <i>Replacement of items that deteriorate gradually</i> and <i>items that fail completely</i> .	CO1	K2

7. Attempt any **one** part of the following:

10 x 1 = 10

(a)	A project has the following activities with predecessors: <table><tr><th>Activity</th><th>Predecessor</th><th>Time (Days)</th></tr><tr><td>A</td><td>-</td><td>23</td></tr><tr><td>B</td><td>-</td><td>8</td></tr><tr><td>C</td><td>-</td><td>20</td></tr><tr><td>D</td><td>A</td><td>16</td></tr><tr><td>E</td><td>A</td><td>24</td></tr><tr><td>F</td><td>B,D</td><td>18</td></tr><tr><td>G</td><td>C</td><td>19</td></tr><tr><td>H</td><td>B</td><td>4</td></tr><tr><td>I</td><td>F,G</td><td>10</td></tr></table> Construct a network diagram for this project. Then calculate earliest start time (EST) and latest start time (LST) for each activity in CPM, Find total project completion time with critical path.	Activity	Predecessor	Time (Days)	A	-	23	B	-	8	C	-	20	D	A	16	E	A	24	F	B,D	18	G	C	19	H	B	4	I	F,G	10	CO1	K2
Activity	Predecessor	Time (Days)																															
A	-	23																															
B	-	8																															
C	-	20																															
D	A	16																															
E	A	24																															
F	B,D	18																															
G	C	19																															
H	B	4																															
I	F,G	10																															
(b)	Explain the PERT method for project analysis. For an activity, three time estimates are: Optimistic time (to) = 4 days; Most likely time (tm) = 6 days; Pessimistic time (tp) = 11 days Calculate the expected time (te) and variance of the activity. Also explain how PERT incorporates uncertainty in project completion time. Differentiate between PERT and CPM.	CO1	K2																														